

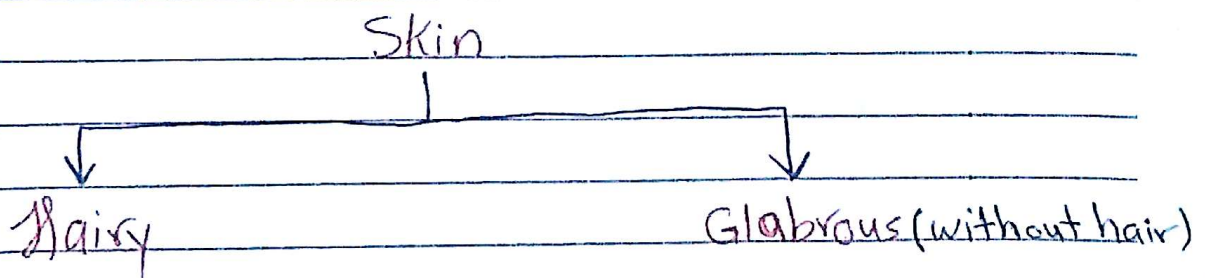
Lab # 12 :- Histology of Human Organs

* Organ → similar tissue that cooperate with each other to carry specific function.

* Histology → The science which study the organs.

[I] Skin :-

- The largest organ of the integumentary system.
- Has multiple layers of ectodermal tissue.
- Guards the underlying muscles, bones, ligaments and internal organs.



* Functions of Skin :-

- | | |
|----------------------------------|----------------------|
| 1- Protection | 7- Excretion |
| 2- Sensation | 8- Absorption |
| 3- Heat regulation | 9- Water resistance. |
| 4- Control of evaporation | |
| 5- Aesthetics and communication. | |
| 6- Storage and synthesis | |

* Skin Layers:-

1- Epidermis:-

- the outermost layer of the skin.
- made up of stratified squamous epithelium with underlying basal lamina.
- Contains no blood vessels, cells in the deepest layers are nourished almost exclusively by diffused oxygen from surrounding air and to a far lesser degree by blood capillaries strata.

* Epidermis divides to 5 sublayers:-

1. Stratum corneum
2. Stratum lucidum (in palms + bottom of feet).
3. Stratum granulosum
4. Stratum spinosum
5. Stratum germinativum (stratum basale).

* Cells are formed through mitosis at the basale layer and the daughter cells move up the strata changing shape and composition.

* The cytoplasm of these cells is released and the protein Keratin is ~~int~~ inserted. They reach the corneum and slough off, this process is called Keratinization.

- * The keratinized layer of skin is responsible for:-
 - keeping water in the body.
 - " other harmful chemicals and pathogens out.
 - making skin a natural barrier to infection.

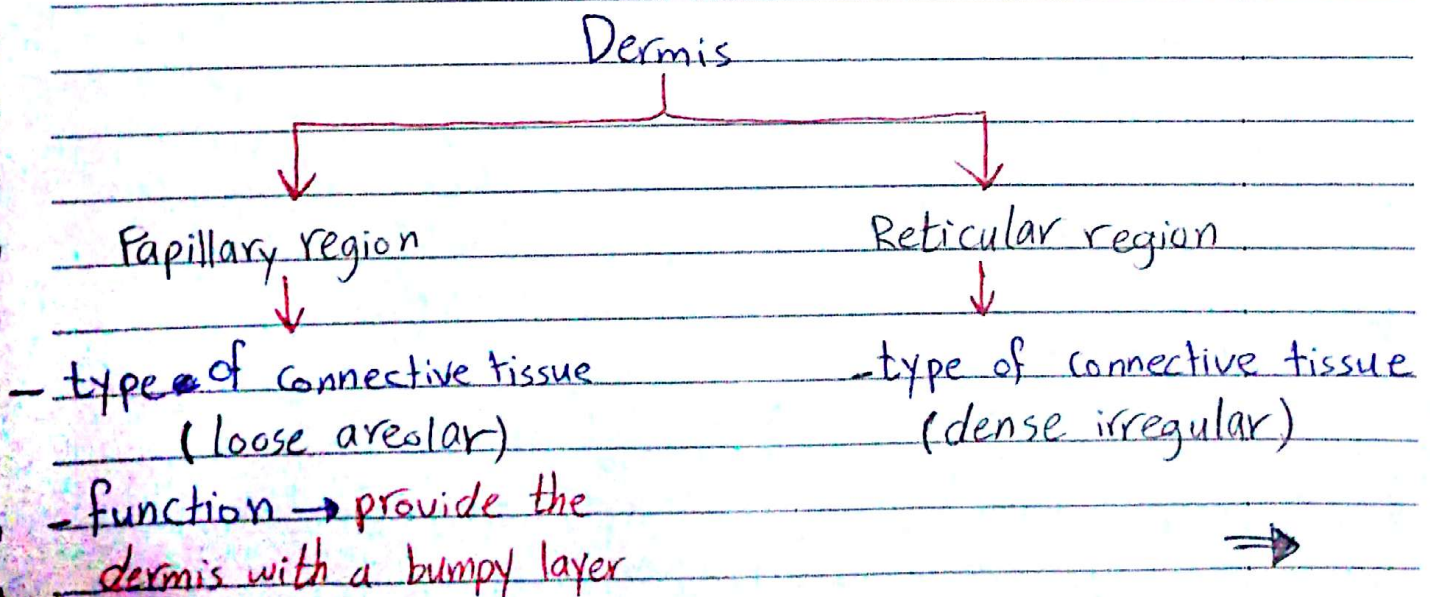
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2- Dermis:-

- beneath the epidermis.
- type of tissue → Connective tissue and is tightly connected to the epidermis by a basement membrane.

* Functions of Dermis:-

- 1- Cushions the body from stress and strain.
- 2- harbors many nerve endings that provide the sense of touch and heat.



- * Bumpy Surface → 1- interdigitates with the epidermis.
2- strengthening the connection between the dermis + epidermis.

* Reticular region :-

- deep to the papillary region and is much thicker.
- Three types of protein fibers :-
 - * collagenous
 - * elastic
 - * reticellular fibers.
- Contains → root of the hair, sweat glands, sebaceous gland, receptors, nails, blood vessels.

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* sebaceous gland produce sebum which naturally heavy skin lubricant.

* Skin becomes heavy & thick in texture when produces excessive sebum.

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⊗ Hypodermis :-

- type of tissues → • Loose CT • elastin
- It's not part of the skin.

- Functions → 1- attach the skin to underlying bone and muscle.
2- supplying the skin with blood vessels and nerve.

* main cell type → fibroblast, macrophages, adipocytes.

[2] Intestines easily,

- vital organs in the gastrointestinal tract of our digestive system.
- Functions → to digest food and enable the nutrients released from that food to enter into the blood stream.

Intestines

Small

Large

smaller in diameter,
much longer and more
massive than the large
intestine.

larger in diameter,
shorter.

* The basic arrangement of layers in the intestinal wall are seen in both the duodenum + jejunum. In each, there is:-

1* Mucosa

2* submucosa

3* muscularis

4* adventitia or a serosa.

* Mucosa:-

- The innermost layer of the intestines.
- finger-like projections called the villi.

* the villi lined by simple columnar epithelium tissue → Contain
 → mucus-secreting goblet cells → ^{fxn} mucous secreting gland.

* Fxn of epithelial cell of the mucosa:

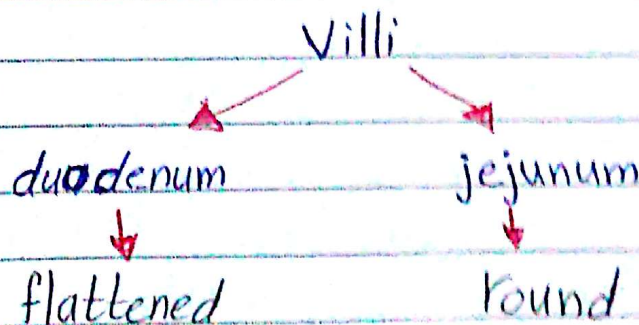
- 1- absorb the nutrients & water from digested food.
- 2- transfer these substances to blood in nearby capillaries.

* Surface of epithelium ^{has} → microvilli → increase the effecting of absorption.

* Nucleus of epithelial cells → ovoid, located in the lower half of the columnar cell Basement membrane.

* Basement membrane → delicate membrane supports the epithelium and composed primarily of reticular fibers embedded in an amorphous protein polysaccharide ground substance.

* Muscularis mucosa → between mucosa and submucosa
 → The lamina propria terminates at it.



* Lamina propria consist of:- (CT $\Rightarrow$ Connective Tissue)

1. Loose CT
2. blood vessels.
3. Lymphatic vessel
4. Smooth muscle fibers
5. Cells of CT

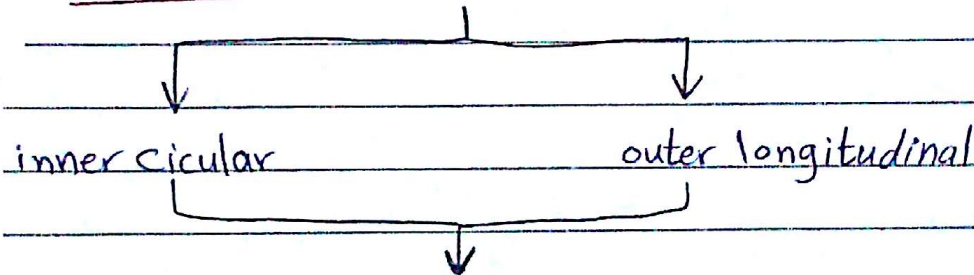
2* Submucosa:-

→ only in the duodenum, not jejunum, contains:-
Connective tissues.

- mucous glands.

* They are a continuation of the pyloric glands found in the stomach*.

3* Muscularis:- two layers



smooth muscles fibers, separated by reticular and collagenous CT containing nerve fibers and parasympathetic ganglion cells.

4* Serosa:-

→ outmost layer of the retroperitoneal part of intestine is the adventitia.

→ Contains of Loose CT.

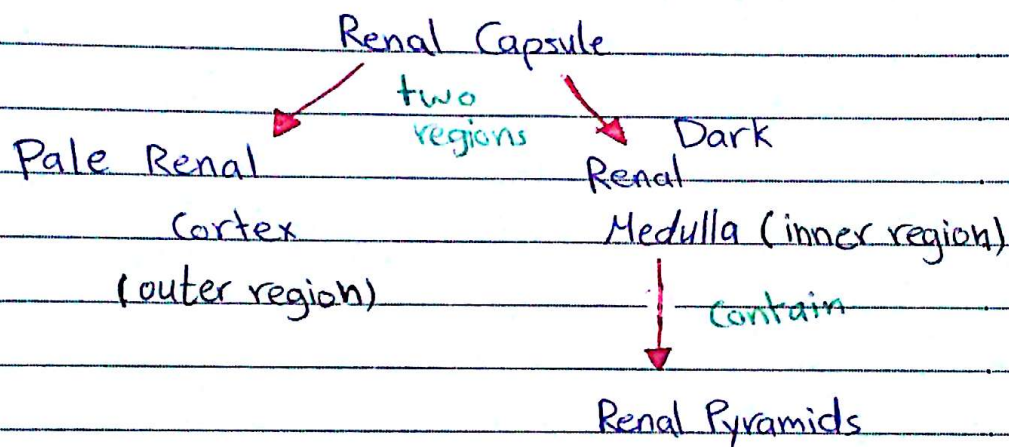
→ main parts of it:- nerves, blood vessels, lymphatic & vessels, mesothelium.

• duodenum → retroperitoneal • jejunum → intraperitoneal

3] Kidney:- الكلى

* Kidney:- two bean-shaped that are located on either side of the body underneath the diaphragm.
الكبد الكلى

* Renal Capsule → smooth, tough fibrous capsule that enclose the kidney.



* Nephrons → tube like structures that carry out the task of filtration of blood.

* Kidney has an adrenal gland at the top of it → they are surrounded by two layers:-

- 1- fat (perirenal + pararenal fat).
- 2- renal fascia.

The Medulla :-

- region lacking renal corpuscles.
- Contains of → a. most of ^{the} length of the loops of Henle.
b. most of the length of the collecting ducts (collecting tubules).
- is formed of pyramidal structures known as renal pyramids.
- The apex of a pyramid is the papilla → meets the calyx.
- ⊗ The spaces between renal pyramids are called renal columns.

The Cortex :-

- area containing renal corpuscles, proximal and distal tubulus, descending thick limbs.
- Types of tissues in Cortex :-
 1. the medullary rays (Henle)
 2. the labyrinth of ~~to~~ ludwing → contains small structures called glomeruli or Malpighian tubules.
- * The space between the cortex and the medulla contains blood vessels → form an arcade shape and run parallel to the surface of the cortex.

The Capsule :-

→ The outer thin capsule contains a relatively large collagen fibers arranged in a two dimensional array.

* The outer thin capsule of dense FECT*
(Fibroelastic Connective Tissue)

→ The outer capsule is covered by mesothelium on ~~series~~ surfaces exposed to the peritoneal cavity.

→ 2 layers:-

1. outer dense FECT → the outer layer of the capsule.

2. myofibroblast layer → " inner layer " " " "

→ myofibroblast layer lies between the dense FECT outer capsule and the parenchyma of the cortex, but may be absent in smaller mammals.

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Renal Pelvis :-

→ Contains of :- 1- major calyx → branches at the base of the renal pelvis.

2- minor calyx → branches away from the renal pelvis.

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Connective Tissues :-

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the material between these parts.

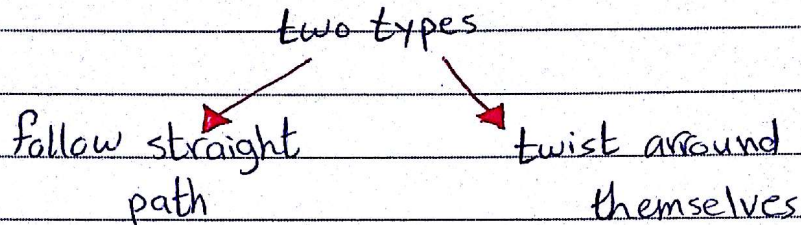
* Parts of kidney :- blood vessels.

② Stroma

③ Collecting tubules.

* Renal Tubules *

→ Small tubes have a diameter of 0.2 mm.



→ every renal tubules is originates from Bowman's capsule ~~and~~ around the glomerulus.

* Nephrons *

→ Functions: Filtration, Reabsorption, Secretion.

* Each kidney contains more than 800,000 nephrons *

→ Consist of: Renal Capsule, Renal Tubules

↓
(قروءة) من الأنبوب

→ Loop of Henle is 3 parts:- descending thick limb.

Thin limb.

- Ascending thick limb.

* قراءة بالتفصيل من
الأنبوب

* من الأنبوب

4) Liver :-

- The largest gland in the body.
- Its functions as two types of glands

Exocrine gland

Endocrine gland

fxn:- secreting bile via a system into the duodenum

fxn:- synthesizing & releasing organic compound into the blood stream

→ Components of Liver:-

- 1- Hepatic lobule → the functional unit of the liver.
- 2- Central vein → forms the axis of hepatic lobule, receives blood from the hepatic sinusoids, drains into intercalated vein.
- 3- Liver cells
- 4- Sinusoids
- 5- Hepatic cords
- 6- Bile canaliculi
- 7- Hepatocyte "the main tissue cell in liver"
- 8- Portal space.

جذع
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[5] Spinal Cord :- كبل الشوكي

(G.M → Grey Matter)



Contains of :-

neurons, glial cells, non-myelinated axon.

→ makes up the central nervous system.

→ five groups of nervous system:-

1. Cervical region → 8 pairs of nerves.

2. Thoracic region → 12 pairs

3. Lumbar → 5 pairs

4. Sacral → 5 pairs

5. Coccygeal → 1 pair.

→ Spinal Cord Contains :- Grey matter & White matter.

forms a sort of H with 3 horns

Contains myelinated axon

* Horns of G.M :-

- ventral horn → somatic motor nuclei

- lateral horn → autonomic " "

- dorsal horn → divided with the farthest back.

* * * * *

[6] Trachea :- القصبة الهوائية

→ begins just under the larynx and runs down behind the breastbone.

→ divides into two smaller tubes called bronchi.

→ Composed of 20 rings of tough cartilage.

→ the back part of each ring is made of muscle and connective tissue.

→ 4 layers of tissues make up the walls of trachea:-

1. Mucosa → the innermost layer consists of ciliated pseudostratified columnar epithelium with many goblet cells. v. imp

* goblet cells → Produce sticky mucus to coat the inner lining of the trachea, catch any debris present in inhaled air before it reaches the lungs.

* Hair-like cilia → On the surface of the columnar cells, beat together to push mucus away from the lungs.

* larynx → expelled during coughing, swallowed and digested in the stomach.

* * * * *

2. Submucosa → contains blood vessels + nervous tissue.

support the outer layers of the tracheal wall.

* Collagen protein fibers, elastic, reticular

↓
give support + elasticity to tracheal wall

* Longitudinal smooth muscle fibers are present in the posterior trachea between the ends of the cartilage rings.

3. Hyaline Cartilage → C-shaped cartilages surrounding the submucosa, forms the supportive rings of the trachea.

→ Consists lacunae where the chondrocyte ^{set} ~~are~~ located.

→ Hyaline provides a strong, yet flexible structure.

maintains an open airway, is resistant to external stresses.

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4. Adventitia → the outermost layer in the trachea

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layer of areolar ct that loosely anchors the trachea to the surrounding soft tissues.

درسوا الرسومات
من المايكروال

* Good Luck ^^ *

* Raneem Jaradat *

* Biotechnology & Genetic Engineering *